



VIT-AP
UNIVERSITY

Capstone Review - 1

**AI-Powered Continuous CCTV Monitoring &
Mobile Alert System for Adults with Autism and
Mental Health Disorders**

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Introduction

This project proposes an AI-powered CCTV monitoring system designed for adults with autism and mental health disorders. The system aims to detect abnormal behavior and notify caregivers in real time while preserving patient privacy. The architecture is informed by a comprehensive review of 40 research papers, focusing on the transition from raw video analysis to skeleton-based action recognition.

Objectives

- Continuous monitoring
- Detect distress, self-harm, agitation, falls, and violence
- Real-time alerts via mobile application
- Privacy preservation (via skeleton-based keypoint extraction)
- Reduced false alarms through temporal filtering

Problem Statement

Traditional observation methods are limited, expensive, and often miss critical behavioural changes. Clinical observation suffers from observer fatigue, and raw video monitoring poses significant privacy and ethical challenges in healthcare settings.

Literature Review

A review of 40 research papers reveals three key strategic themes:

1. **Raw Video vs. Skeleton-Based Recognition:** Modern frameworks favor pose estimation (e.g., MediaPipe) combined with Spatial-Temporal Graph Convolutional Networks (ST-GCN) over 3D-CNNs on raw RGB frames. This approach provides significant speedups, is resilient to lighting variations, and ensures strict privacy adherence by discarding raw footage.
2. **Psychiatric Domain Specificity:** Generic models often fail on subtle behavioral precursors (hand-flapping, head-banging). High-performing models incorporate scene-motion context and 24-hour spatial trajectory tracking.

3. **Edge Processing:** Lightweight local inference (converting frames to coordinate vectors) paired with cloud push services (e.g., Firebase) is essential to achieve near-zero alert latency.

Proposed Methodology

- Phase 1: Data preparation (Skeleton extraction from simulated/clinical video).
- Phase 2: Model training (ST-GCN training on coordinate vector sequences in the cloud).
- Phase 3: Edge inference (OpenCV/MediaPipe running locally on CPU to extract joints).
- Phase 4: Cloud integration (JSON anomaly payload triggers AWS backend).
- Phase 5: Mobile application (Flutter-based notification delivery via Firebase).

Architecture

CCTV → Pose Extraction → ST-GCN → Temporal Filtering → Cloud Backend → Mobile Alerts

Models and Technologies

- **Models:** ST-GCN, CNN-LSTM, MediaPipe, YOLOv8-Pose.
- **Technologies:** PyTorch, TensorFlow, AWS, Flutter, Firebase.

Expected Outcomes

Continuous monitoring, faster response times, improved patient safety, and significantly reduced emergency response latency.

Future Scope

Implementation of federated learning for privacy-preserving model updates, integration with broader smart-home environmental sensors, and large-scale clinical deployment.